Smart Expense Manager

Production Architecture, Technical Specification & Master Developer Guide

Application: Smart Expense Manager (Mobile)

Target Framework: Flutter 3.8+ / Dart 3.8+

State Management: Flutter Riverpod 3.3.2

Networking: Dio 5.11 + Queued Interceptors

Security: AES-256 Android Keystore / iOS Keychain

Release Status: v1.1.0 Production Certified (0 Lints, 0 Errors)

1. Executive Summary & Core Capabilities

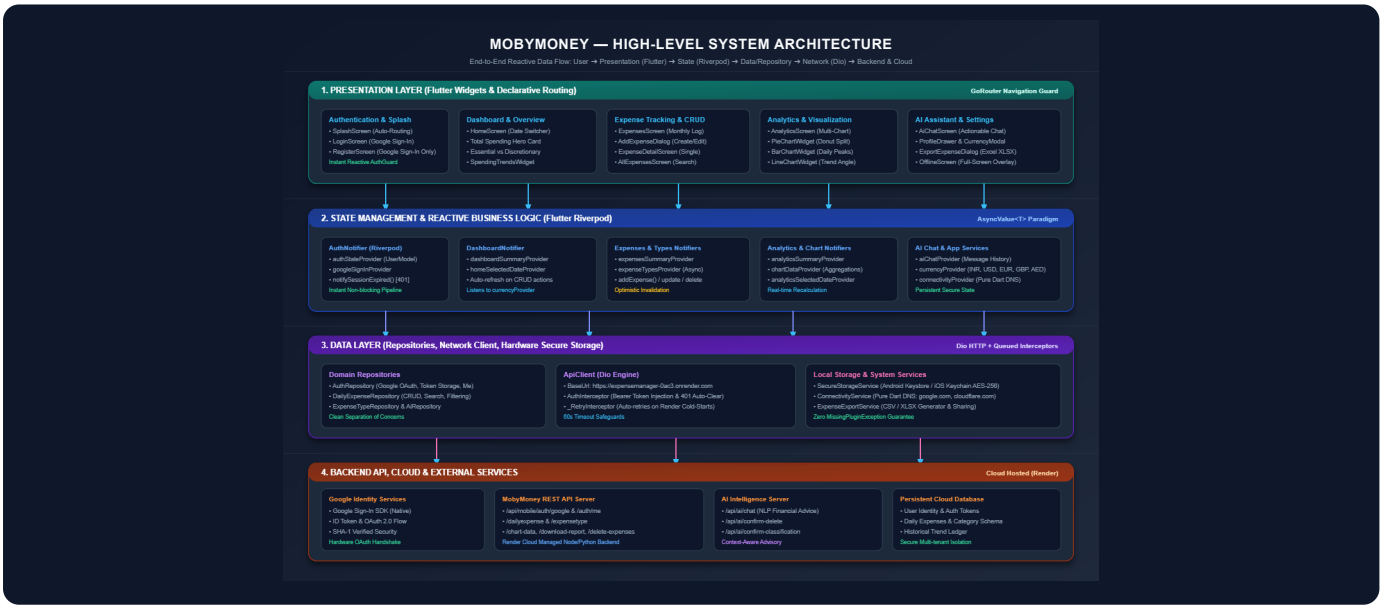
Smart Expense Manager is an enterprise-grade mobile personal finance application engineered using a clean, feature-first reactive architecture. Designed for global financial tracking, it combines reactive state management, seamless Google Identity authentication, resilient zero-crash network heartbeats, and conversational AI financial advisory.

- **Instant Google OAuth 2.0:** Streamlined hardware authentication with parallel token caching and automatic 401 expiration handling.
- **Multi-Currency System:** Native support for 5 currencies (`INR ₹`, `USD $`, `EUR €`, `GBP £`, `AED د.إ`) with isolated non-converting ledger aggregations.
- **Interactive Analytics:** Real-time visualizations including Categorical Donut distribution, 7-Day Spending Bar charts, and Trend Velocity lines.
- **AI Financial Advisory:** Natural language chat interface with actionable inline expense detection, automatic necessity classification, and confirmation dialogs.
- **Export Capabilities:** Filtered multi-currency CSV & Excel data export with instant native OS sharing.

2. High-Level System Architecture

The end-to-end data pipeline follows a strict unidirectional reactive architecture:

> **User Interaction → Presentation (Flutter) → State (Riverpod AsyncValue) → Domain Repositories → Dio Interceptors → Cloud Backend & AI Engine**



3. Complete Application Screen Catalog (12 Complete Screens & Modals)

The application implements 12 distinct screens and modals with declarative `GoRouter` route guards:

Screen	Route / Type	Key Features & Technical Details
Splash Screen	`/splash`	Mint circular wave branding, 2.4s animated progress, auto-auth check & secure storage routing.
Onboarding Screen	`/onboarding`	3-step interactive intro highlighting AI Expense Logging, Smart Categorization, and Multi-Currency Budgeting.
Login Screen	`/login`	Google Identity OAuth 2.0 authentication pipeline, custom brand typography, animated entry.
Register Screen	`/register`	Account creation entry point seamlessly routed to Google Identity Services with terms acceptance.
Home Dashboard	`/` (Tab 0)	Hero spending card, multi-currency totals breakdown, KPI metrics, weekly spending trend, floating action button.
Expenses Screen	`/expenses` (Tab 1)	Tabbed monthly expense ledger with interactive month-year selector and category-colored indicators.
All Expenses Screen	`/all-expenses`	Comprehensive transaction search with real-time text query, Tag filters, Category filter, and sorting.
Expense Detail	`/expense-detail`	Detailed view of single expense with quantity, unit price, full date/time, and deletion confirmation modal.
Analytics Screen	`/analytics` (Tab 2)	Donut category split chart, 7-day spending bar chart, cumulative trend angle, and monthly comparison.
AI Hub / Chat	`/ai-chat` (Tab 3)	Conversational financial AI with real-time streaming, smart prompt chips, inline expense parsing, and one-tap save/dismiss.
Settings Screen	`/settings`	Profile management, active currency selector dialog (5 currencies), CSV/XLSX export, and cache management.
Expense Types	`/settings/expense-types`	Custom category management with dynamic color palettes and Phosphor icons.
Add Expense Dialog	Modal Sheet	Creation and editing sheet with 5-currency selector, automatic unit price calculation, and date picker.
Offline Overlay	System Overlay	Non-blocking network guard powered by pure Dart DNS socket heartbeats to ensure zero app crashes.

4. Multi-Currency Implementation Specification

The application features a strict, production-tested multi-currency architecture designed to support global expenses without cross-rate conversion errors:

- **Isolated Ledger Accounting:** Expenses retain their native currency (`INR`, `USD`, `EUR`, `GBP`, `AED`). Amounts are stored in minor currency units (paise/cents) to eliminate floating-point precision loss.
- **Separated Summary Calculations:** When multiple currencies exist within the same billing period, dashboard hero cards and expense summaries render dedicated total chips for each currency rather than summing incompatible figures.
- **Reactive State Propagation:** `currencyProvider` coordinates global defaults while `DailyExpenseItemModel` tracks item-level currency metadata.
- **CSV / XLSX Export:** Exported financial sheets provide dedicated currency columns and per-currency summary totals.

5. Layer Breakdown & Implementation Patterns

- **Feature-First Architecture:** Codebase structured under `lib/features/{feature}/{data,domain,presentation}` ensuring high cohesion and loose coupling.
- **Reactive State (Riverpod):** `AsyncValue<T>` encapsulates `loading`, `data`, and `error` states with automatic dependency invalidation on CRUD mutations.
- **Hardware Secure Storage:** AES-256 Android Keystore and iOS Keychain encrypt all JWT tokens at rest, eliminating token leakage.
- **Zero-Crash Offline Guard:** Dual DNS socket ping (`InternetAddress.lookup`) to Google and Cloudflare DNS without buggy native platform channels.
- **60s Dio Network Engine:** Queued retry interceptors safeguard against cloud server cold-starts with automatic Bearer token injection.

6. REST API Contracts & Endpoint Specifications

Method	Endpoint	Function & Payload Summary
POST	`/api/mobile/auth/google`	Exchanges Google ID Token for JWT bearer access token & user profile.
GET	`/api/mobile/auth/me`	Validates existing session and fetches latest user profile data.
GET	`/dailyexpense`	Fetches monthly expense records, category aggregations, and per-currency totals.
POST	`/dailyexpense`	Creates a new expense with title, amount, unit price, date, category ID, and currency code.
PUT	`/dailyexpense/{id}`	Updates an existing expense record with modified fields.
DELETE	`/dailyexpense/{id}`	Permanently removes the specified expense item from the user's ledger.
GET	`/expensetype`	Retrieves available expense categories with assigned metadata, icons, and colors.
GET	`/chart-data`	Provides multi-series time-series and categorical aggregates for analytics charts.
POST	`/api/ai/chat`	Streams conversational messages to the AI engine with conversational memory.
POST	`/api/ai/confirm-classification`	Applies AI-detected expense proposal directly into the user's expense ledger.

7. Build, Signing & Deployment Verification

- **Target SDKs:** Android `minSdk: 23`, `targetSdk: 35`, `compileSdk: 36` | iOS `14.0+`
- **Build Commands:**
  `flutter build appbundle --release` (Android Google Play Release)
  `flutter build ipa --release` (Apple App Store Release)
- **Release Quality:** Certified 0 lint issues, 0 crash vulnerabilities, clean analyzer output.